

THE CHAISE LONGUE THEOREM — LEAN CERTIFICATE

A machine-checked proof, in Lean 4 with Mathlib, of Main Theorem': the algebraic core of Conjecture 1.2 of Degtyarev–Shimada for every odd degree

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0. Verdict

For every odd $m \geq 1$ and every $k \geq 0$, Lean 4 proves:

- the ring $R = \mathbb{Z}[t_1, \dots, t_{2k+1}]/(t_1^m - 1, \dots, t_{2k+1}^m - 1)$ modulo the ideal generated by the $(2k+1)!!$ elements ψ_j of Degtyarev–Shimada is a **free abelian group of rank $m^{2k+1} - Q_k(m)$** ;
- for **every field** F , $\dim_F F[G]/(\psi_j) = m^{2k+1} - Q_k(m)$.

This is Main Theorem' of the paper, stated there for $m \geq 3$ and $k \geq 1$. It holds for every prime p at once, including every prime dividing m .

The whole proof compiles on the author's Mac. It contains no `sorry`, no `admit`, no `native_decide` and no added axiom. The final theorem depends only on the three standard axioms of Lean and Mathlib: `propext`, `Classical.choice` and `Quot.sound`.

Main Theorem' is not the Main Theorem. The Main Theorem is Conjecture 1.2 itself: $H_{2k}(X; \mathbb{Z})/L(X)$ is torsion free. It follows from Main Theorem' by one topological result, [DS, Theorem 1.1(a)]: the torsion of $R/(\psi_j)$ equals the torsion of $H_{2k}(X; \mathbb{Z})/L(X)$. That result rests on Pham's description of the homology of Fermat varieties and on the intersection numbers [DS, Theorem 2.2]. **It is not formalized; it is cited.** So:

Lean certifies the algebraic statement (Main Theorem') completely. The Main Theorem follows from it by one cited topological result, the same one the paper cites.

§1 states exactly what is proved. §2 states exactly what is not. §3–§8 describe how the result was obtained and how to check it.

1. The certified statement

1.1 The theorem, verbatim

Printed by Lean with `#check (log proyectos_lean/check_run29.log)`:

```
@ColAssembly.mainTheorem' : ∀ {m : ℕ},
  Odd m →
  ∀ (k : ℕ),
  (Module.Free ℤ (RZ (2 * k + 1) m / IZ m k) ∧
   Module.finrank ℤ (RZ (2 * k + 1) m / IZ m k) = m ^ (2 * k + 1) - TheoremB.Qk k m) ∧
  ∀ (F : Type) [inst : Field F],
  Module.finrank F (ColSplit.GA F (2 * k + 1) m / ColUpper.IK F m k) = m ^ (2 * k + 1) -
  TheoremB.Qk k m
```

'ColAssembly.mainTheorem'' depends on axioms: [propext, Classical.choice, Quot.sound]

The integral part and the field part are also stated as two separate theorems. They are `ColAssembly.mainTheorem'_Z` and `ColAssembly.mainTheorem'_field`. The field version quantifies over fields in every universe; the combined theorem uses $F : \text{Type}$.

1.2 The definitions it uses, verbatim

These are all the definitions needed to read the statement. They are printed by `#print` in the same log.

```
def ColAssembly.idealZ : (d : ℕ) → ℕ → Ideal (MvPolynomial (Fin d) ℤ) :=
  fun d m => Ideal.span (Set.range fun i => MvPolynomial.X i ^ m - 1)

@[reducible] def ColAssembly.RZ : ℕ → ℕ → Type :=
  fun d m => MvPolynomial (Fin d) ℤ / idealZ d m

def ColAssembly.IZ : (m k : ℕ) → Ideal (RZ (2 * k + 1) m) :=
  fun m k => Ideal.span (Set.range fun J => (Ideal.Quotient.mk (idealZ (2 * k + 1) m)) (ColUpper.psiP
  ℤ m J))

def ColUpper.psiP : (R : Type u_1) → [CommRing R] → ℕ → {k : ℕ} → BallotBound.Matching k →
MvPolynomial (Fin (2 * k + 1)) R :=
  fun R m {k} J =>
    (∏ a with a < ↑J a, (ColUpper.tP R k (↑J a) - 1)) *
    ∏ a with 0 < a ∧ a < ↑J a, ColSurv.phi m (ColUpper.tP R k a * ColUpper.tP R k (↑J a))

-- tP R k : Fin (2k+2) → MvPolynomial (Fin (2k+1)) R is Fin.cases 0 X (t_0 := 0, t_a := X_{a-1} for
a ≥ 1)

def ColSurv.phi : {A : Type u_2} → [Semiring A] → ℕ → A → A :=
  fun m u => ∑ s ∈ Finset.range m, u ^ s

def BallotBound.Matching : ℕ → Type :=
  fun k => { J // ∀ (x : Fin (2 * k + 2)), J x ≠ x ∧ J (J x) = x }

@[reducible] def ColSplit.GA : (F : Type u_1) → [Field F] → ℕ → ℕ → Type u_1 :=
  fun F d m => MvPolynomial (Fin d) F / ColSplit.gaIdeal F d m
  -- gaIdeal F d m = Ideal.span (Set.range fun i => X i ^ m - 1)

def ColUpper.IK : (K : Type u_1) → [Field K] → (m k : ℕ) → Ideal (ColSplit.GA K (2 * k + 1) m) :=
  fun K m k => Ideal.span (Set.range (ColUpper.psiG K m))
  -- psiG K m J is the same formula as psiP, read in GA K (2k+1) m

def TheoremB.Qk : ℕ → ℕ → ℕ :=
  fun k q =>
    ∑ b ∈ (Fintype.piFinset fun _ : Fin ((q - 1) / 2) => Finset.range (2 * k + 3)).filter
      (fun b => 2 * ∑ u, b u = 2 * k + 2),
    (2 * k + 2).factorial / ∏ u, (b u).factorial ^ 2
```

1.3 Faithfulness to the paper, line by line

The paper (v7, §1.1–§1.2 and §2) uses the notation of [DS] (arXiv:1405.4683v3).

Paper	Lean	Agreement
$\mathcal{J}$: partitions of $\{0, \dots, 2k+1\}$ into pairs, written $[[j_p, k_i]]$ with $j_i < k_i, j_0 = 0$; $ \mathcal{J} = (2k+1)!!$	Matching k : fixed-point-free involutions of $\text{Fin } (2k+2)$	The same objects. A pair $\{a, J a\}$ with $a < J a$ is $[j_p, k_i]$.
$R = \mathbb{Z}[t_1, \dots, t_{2k+1}]/(t_i^m - 1)$, the group ring of $(\mathbb{Z}/m)^{2k+1}$	$RZ(2k+1) m = \mathbb{Z}[X_0, \dots, X_{2k}]/(X_i^m - 1)$	Identical, with 0-based indices: $t_a = X_{a-1}$.
$\varphi(u) = u^{m-1} + \dots + u + 1$	$\text{phi } m u = \sum_{s < m} u^s$	Identical.
$\tau_J = \prod_{i=0}^k (t_{k_i} - 1)$	$\prod_{a < J a} (t_{J a} - 1)$	One factor per pair, for the larger index $k_i = J a$.
$\psi_J = \tau_J \cdot \prod_{i=1}^k \varphi(t_{j_i} t_{k_i})$	$\cdot \prod_{0 < a < J a} \varphi(t_a \cdot t_{J a})$	The pairs with $j_i \neq 0$, i.e. $i \geq 1$.
t_0 never occurs	$\text{tP } R \text{ k } 0 = 0$ is a placeholder	It is never used: a factor $t_{J a} - 1$ has $J a > a \geq 0$, and a factor $\varphi(t_a t_{J a})$ has $a > 0$.
the ideal $(\psi_J : J \in \mathcal{J})$ of R	$IZ m k$: generated by the images of the $\text{psiP } \mathbb{Z} m J$	Identical.
$F_p[G] = F_p[t]/(t_i^m - 1)$ and $\bar{\psi}_J$	$GA F(2k+1) m$ and $IK F m k$, for every field F	Contains the paper's F_p as the case $F = \mathbb{Z} \text{Mod } p$.
$Q_k(q) = N! \cdot [x^N] I_0(2x)^{(q-1)/2}$, $N = 2k+2$, $I_0(2x) = \sum x^{2b}/(b!)^2$	$\sum \{b : \text{Fin}((q-1)/2) \rightarrow \mathbb{N}, 2\sum b = 2k+2\} (2k+2)! / \prod (b_u!)^2$	The coefficient of x^N in $\prod_u \sum_b x^{2b}/(b!)^2$, times $N!$. Each quotient is an integer: it is a multinomial coefficient with parts b_u . So the division in $\mathbb{N}$ is exact. The bound $b_u \leq 2k+2$ is automatic, since $b_u \leq k+1$.
«free abelian group of rank r »	$\text{Module.Free } \mathbb{Z} M \wedge \text{Module.finrank } \mathbb{Z} M = r$	Here $r \geq 1$ (see below), so $\text{finrank} = r$ forces M finitely generated. The statement therefore says $M \cong \mathbb{Z}^r$.
$m^{2k+1} - Q_k(m)$	$m(2k+1) - Q_k m$ in $\mathbb{N}$ (truncated subtraction)	Never truncates: the field part shows $Q_k m = \dim_F I_F \leq \dim_F F[G] = m^{2k+1}$. In fact $Q_k(m) = I \leq (m-1)^{2k+1} < m^{2k+1}$, so $r \geq 1$.

Three further checks tie Q_k to its meaning:

- `TheoremB.card_closedTuples`: $Q_k k q$ is the number of closed tuples in T^{2k+2} , for every point set T with $|T| = q - 1$. This is Lemma 2.2 of the paper.
- `ColUpper.card_Gamma`: $Q_k k m = |\Gamma_{\mathcal{J}}|$, the set of [DS, Definition 1.3], over a field with m distinct m -th roots of unity.
- `TheoremB/Checks.lean` evaluates Q_k by decide at the paper's table: $Q_0(3) = 2$, $Q_0(5) = 4$, $Q_1(3) = 6$, $Q_1(5) = 36$, $Q_1(7) = 90$, $Q_1(9) = 168$, $Q_2(3) = 20$, $Q_2(5) = 400$, $Q_3(3) = 70$, and $Q_k(3) = C(2k+2, k+1)$ for $k \leq 4$.

Generality. The paper states Main Theorem' for $m \geq 3$ and $k \geq 1$. Lean proves it for every odd $m \geq 1$ and every $k \geq 0$, and the field part for every field, not only F_p . The degenerate cases are consistent:

- at $m = 1$, $R = \mathbb{Z}$, $Q_k(1) = 0$, and the rank is 1;
- at $k = 0$, the statement is the paper's trivial case.

1.4 The other main theorems of the last piece

Also printed verbatim in `check_run29.log`, each with axioms `[propext, Classical.choice, Quot.sound]`:

- `ColAssembly.prop69` (Proposition 6.9, the colour reduction): for a field F of characteristic $p \neq 2$ containing r distinct r -th roots of unity, r odd, $m = p^v \cdot r$: $\dim_F I_F = Q_k(m)$.

- `ColAssembly.theorem_every_field`: for odd m and every field F , $\text{rank}_F(A) = Q_k(m)$ and $\dim_F I_F = Q_k(m)$. Here $A = \text{ColUpper.intMatU } m \ k$ is the integer matrix of the $t^\nu \psi_j$ in the monomial basis.
 - `ColAssembly.rank_map_algebraMap`: the rank of a matrix does not change under an extension of fields.
-

2. What is not certified

1. **The topology.** Theorem 0(a), (c), (d) of the paper, i.e. [DS, Theorems 1.1(a) and 1.4], rest on Pham's theorem [Ph] and [DS, Theorem 2.2]. They identify the torsion of $R/(\psi_j)$ with that of $H_{2k}(X; Z)/L(X)$ and give the rank of $L(X)$. **Not formalized. Mathlib has no singular homology of complex Fermat varieties.**
 2. **Hence the Main Theorem itself** — Conjecture 1.2 as a statement about homology — is certified only in its algebraic form.
 3. **Not formalized, and not on the path of the proof:**
 - Corollary H (the integral Hodge conjecture);
 - Corollary W (partial Fermat varieties);
 - Theorem A of §8 and the counterexample of §9;
 - §10 and §11;
 - the verification record of §12.
 - *Formalized although not needed:* Theorems 5.9 and 5.9' (equality over every field, freeness of C/V_Λ over Z) and the $q = 3$ case of §4.
 4. **Faithfulness is an audit, not a theorem.** Lean guarantees that the proof is correct for the statement as written. That the Lean statement says what the paper says is checked in §1.3 by the author's audit. **A reader should check §1.2–§1.3 independently: about a dozen short definitions.** This is the one place where trust in the author remains.
 5. **Timing data.** Aristotle's own start and end times per task are shown on its web page but were not exported. The per-piece times in §6 are the download times of the result files: an upper bound for completion, not for Aristotle's work.
-

3. How the proof is organised in Lean

The paper's proof of Main Theorem' (§5–§7, Corollary 7.7) was cut into 29 pieces. Each piece is one short self-contained Markdown file `q_*.md`, written from the paper with statement, proof and checks.

The 29 pieces produced 29 folders under `RequestProject/`, one per piece. Pieces 1 and 2 are the pilot, and piece 1 also produced `RequestProject/Main.lean`. No piece ever modified a file of an earlier piece: every tarball was compared file by file with the compiled project, and all earlier `.lean` files were byte-identical.

#	Source file	Paper	Lean folder	Main Lean results
1	q3_ballot_lower_bound.md	§4, $q = 3$, lower bound	Ballot, Main.lean	BallotBound.finrank_V_ge, card_ballotSets
2	q3_upper_bound.md	§4, upper bound and equality	Upper	BallotBound.finrank_V_le, $\text{finrank}_{V \leq q}$
3	q_peeling_lemma.md	Lemma 5.1	Peel	Peel.W_isIdeal, $W_{m,ono}$, finrank_eq_sum, card_eq_sum_Zgt
4	q_chain_lemma.md	Lemma 5.5	Chain	opt_le_opt_succ, opt_mem_initial_segment, S_subE
5	q_P2_monotone_options.md	Prop. 5.6 (P2)	Monotone	opt_weakDom_opt, card_opt_mem_le
6	q_P1_fibres.md	§5.4 (P1)	Fibres	multiset_resPart_consTuple_eq_opt, card_resPart_consTuple_mem
7	q_P3_identities.md	§5.2, Lemma 5.7	Tight	coeffY_D, sum_eps_Delta, map_relabel_VLam
8	q_P3_lifts.md	Prop. 5.8 (P3)	Lifts	Lifts.layer_subset_W
9	q_induction.md	Theorem 5.3	Induction	Induction.finrank_VLamAll_ge_card_ZLam
10	q_theorem_B_lower.md	Lemmas 5.2, 2.2, 3.1; Theorem B ($\geq$)	TheoremB	finrank_DIdeal_ge, card_closedTuples, card_Gamma'
11	q_degeneration.md	Lemma 5.8', degeneration	Degeneration	finrank_genIdeal_le_card_supportSet
12	q_support.md	Theorem 5.9	Support	Support.finrank_VLamAll_eq_card_ZLam
13	q_every_field.md	Theorem 5.9' (fields)	EveryField	finrank_VLamAll_eq_card_ZLam, finrank_DIdeal_eq, rank0ver_le_rank0ver_rat
14	q_free_Z.md	Theorem 5.9' (over Z)	FreeZ	free_quotient_VZ, quotient_free_of_finrank_image
15	q_bip_setting.md	§7.1–§7.2, Lemmas 7.1, 7.2	Bip	finrank_eq_sum_Wslice, lemma72_i/ii/iii
16	q_bip_options.md	§7.3, Lemma 7.3	BipOpt	lemma73_i ... lemma73_v
17	q_bip_P2.md	Prop. 7.4	BipP2	Bip.prop74_i, prop74_ii
18	q_bip_P3.md	Prop. 7.5	BipP3	BipP3.prop75
19	q_bip_induction.md	Lemma S, Theorem 7.6, Theorem C	BipInd	Bip.thm76, Bip.theoremC
20	q_col_splitting.md	Lemma 6.1	ColSplit	lemma61_i, lemma61_ii, lemma61_iii
21	q_col_survivors.md	Lemma 6.2	ColSurv	lemma62_i/ii/iii

#	Source file	Paper	Lean folder	Main Lean results
22	q_col_compatible.md	Lemma 6.3	ColComp	lemma63_i ... lemma63_v
23	q_col_tensor.md	Lemma 6.4	ColTensor	lemma64_i ... lemma64_v
24	q_col_decomp.md	Prop. 6.5	ColDecomp	prop65_i/ii/iii
25	q_col_pairs.md	Lemma 6.7	ColPairs	lemma67_i ... lemma67_vi
26	q_col_one.md	Lemmas 2.3, 2.4, Prop. 2.5 (local form), Lemma 6.6	ColOne	lemma23_i, lemma24_i/ii, lemma25_i/ii, lemma66_i/ii/iii
27	q_col_count.md	Lemma 6.8	ColCount	lemma68_i/ii/iii
28	q_col_upper.md	Theorem 0(b), the upper bound (6.1) = [DS, Claim 4.3]	ColUpper	finrank_IK_le, finrank_IK_eq_card_Gamma, card_Gamma, rankOver_rat_intMatU
29	q_col_assembly.md	Prop. 6.9, every field, Main Theorem' (Corollary 7.7)	ColAssembly	prop69, theorem_every_field, mainTheorem'

3.1 The chain of the final theorem

This is how mainTheorem' is proved in Lean. Every arrow is a Lean proof.

1. Upper bound, every field (piece 28, finrank_IK_le): $\dim_F I_F \leq Q_k(m)$.

- $\dim_F I_F = \text{rank}_F(A)$ for an integer matrix A (part (i)).
- $\text{rank}_F(A) \leq \text{rank}_{\mathbb{Q}}(A)$ (piece 13).
- $\text{rank}_{\mathbb{Q}}(A) = \text{rank}_{\mathbb{C}}(A)$.
- $\text{rank}_{\mathbb{C}}(A) = |I|$ by evaluation at the m -th roots of unity: Lagrange interpolation and the Chinese remainder theorem (parts (ii)–(iii)).
- $|I| = Q_k(m)$ via closed tuples (Lemma 2.2, piece 10).

This is [DS, Claim 4.3], re-proved without topology.

1. Lower bound when $p \mid m$ (piece 29, prop69_lower).

- $\dim_F I = \sum_c \dim_F \pi_c(I)$ over the colourings c (Lemma 6.1).
- Each term is $\dim I_{c,1} \cdot \prod_{\zeta} \dim I_{c,\zeta}$ or 0 (Prop. 6.5, Lemma 6.3).
- $\dim I_{c,1} \geq N_1(c)$ (Lemma 6.6, via the §2 translation and Theorem B / Theorem 5.3).
- $\dim I_{c,\zeta} \geq N_{\zeta}(c)$ (Lemma 6.7, via Theorem C of §7).
- $\sum_c N_1(c) \prod_{\zeta} N_{\zeta}(c) = Q_k(m)$ (Lemma 6.8).

2. Every field (theorem_every_field). $\text{rank}_F(A)$ depends only on the prime field (rank_map_algebra-Map).

- Characteristic 0: step 1 gives equality.
- Characteristic $p \nmid m$: over $\overline{F_p}$ the count $|I| = Q_k(m)$ gives equality.

- Characteristic $p \mid m$: $m = p^v r$, r odd, and over $\overline{F_p}$ steps 1 and 2 give equality.
3. **Freeness over $\mathbb{Z}$** (`mainTheorem'_Z`). R has the monomial $\mathbb{Z}$ -basis, and $I_{\mathbb{Z}}$ is the $\mathbb{Z}$ -span L of the rows of A . Since $\text{rank}_{F_p}(A) = \text{rank}_{\mathbb{Q}}(A) = Q_k(m)$ for every prime p , the lemma of piece 14 gives $Z^n/L \cong Z^n -_{Q_k(m)}$.

3.2 The dependency cone

A short Lean program (`checks/Deps29.lean`, `log logs/deps_mainTheorem.log`) collects every declaration that `ColAssembly.mainTheorem'` uses, directly or indirectly, and keeps those defined in the project.

- **Size: 2 367** declarations of the project, in **90** of its 113 files and **26** of its 29 folders. Every one of them is checked by the kernel as part of `mainTheorem'`.
- **Folders the final theorem does not use:**
 - Upper (piece 2, the $q = 3$ upper bound of §4);
 - Degeneration (piece 11);
 - Support (piece 12, Theorem 5.9).

These are exactly the results the paper says are not on the path of the proof: §4, Theorems 5.9 and 5.9' (paper §1.2).

- **Folders the final theorem uses only in part:**
 - piece 1: only the definition of matchings;
 - piece 13: only the rank lemmas of `EveryField/Rank.lean`;
 - piece 14: only the integer-matrix lemma of `FreeZ/Lemma.lean`.

The count per file is in the log. So the chain of §3.1 is not only the author's description: it is what the Lean terms actually use.

4. Trust base

- **What must be trusted:**
 - the Lean 4 kernel, toolchain `leanprover/lean4:v4.28.0`;
 - Mathlib at commit `8f9d9cff6bd728b17a24e163c9402775d9e6a365` (tag `v4.28.0`, `lake-manifest.json`);
 - the three standard axioms `propext`, `Classical.choice`, `Quot.sound`;
 - the audit of the statement (§1.3).
- **What need not be trusted:** Aristotle (Harmonic), the AI system that wrote the Lean proofs, and the author's brute-force checks. Every file Aristotle returned was compiled again on the author's machine, and the kernel re-checks every proof.
- **Forbidden constructs, searched in all 113 files:** `grep` finds no `sorry`, `admit`, `axiom declaration`, `native_decide`, `@[extern]`, `implemented_by`, `unsafe`, `opaque` or `partial def`.
- **Tactic `decide`** occurs in 11 files. Most uses are small finite checks such as the table of Q_k values. It is kernel-checked and adds no axiom, as `#print axioms` confirms.
- **Options.** The files set:
 - resource limits: `maxHeartbeats`, `synthInstance.maxHeartbeats`, `maxRecDepth`, `synthInstance.maxSize`;

- printing options: `pp.*;`
- elaboration options: `autoImplicit false, relaxedAutoImplicit false, grind.warning false.`

None of them affects soundness.

- **Axioms of every main theorem**, printed after each piece (logs `proyecto_lean/check_run*.log` and the audit log): always `[propext, Classical.choice, Quot.sound]`, except `ColTensor.lemma64_v`, which uses only `[propext, Quot.sound]`.
- **Aristotle's own register** for the project lists 137 properties PROVED, 0 NEGATED, 0 in progress.

5. The procedure

For each of the 29 pieces:

1. **Write the piece from the paper.** A self-contained Markdown file with Setting, Statement, Proof and Checks. It names the earlier Lean results it may reuse and forbids changing any existing file or statement.
2. **Brute-force check before sending.** A Python program checks the statement in small cases, with at least one negative control where possible (a deliberately wrong variant that must fail). The numbers are in the table of §6.
3. **Send to Aristotle** (project `d1c5d0ac-9ffc-477b-ae34-0729a44d985c`), one task per piece.
4. **Download** the resulting project (a `.tar.gz`) and extract it into `run<N>/`.
5. **Diff** against the locally compiled project. Stop if any earlier `.lean` file changed. None ever did.
6. **Grep** the new folder for the forbidden constructs of §4.
7. **Build locally:** `lake build RequestProject.<Folder>.Main under /usr/bin/time -l, log build_run<N>.log.`
8. **Print** the main statements (`#check`), the definitions (`#print`) and the axioms (`#print axioms`), log `check_run<N>.log.`
9. **Audit faithfulness:** compare the printed statements and definitions with the piece and with the paper. Accept generalizations; stop on any weakening.
10. **Record** the run in the audit log `INFORME_ARISTOTLE_v1.md` and the state file `ESTADO_ARISTOTLE.md`.

Stop conditions, fixed in advance:

- any property NEGATED by Aristotle;
- a statement that is not faithful;
- a build failure;
- a sorry or an extra axiom;
- a change to an old file.

None occurred.

Twice (pieces 15 and 20) Aristotle's page showed a «review suggested» badge. Both times it accompanied Aristotle's own list of encoding choices (for example, bundling the Setting into a structure). The choices were reviewed and found sound. Several pieces were proved in a more general form than asked:

- unused hypotheses dropped (for example $m \geq 3, k \geq 1, v \geq 1$);
- statements for any field or any ring;

- a different proof of the same statement (for example Mathlib's combinatorial Nullstellensatz in place of the Chinese remainder theorem in piece 2).

Each is noted in the audit log.

6. Data per piece

- **Tarball, md5:** the project returned by Aristotle, file in ~/Downloads, first 8 hex digits of its md5.
- **Downloaded:** the local time the file was saved (CEST). This bounds the completion of the Aristotle task.
- **Build, peak RSS:** the local *lake build* of the new folder (wall-clock seconds, and «maximum resident set size» from `/usr/bin/time -l`).
- **Files / lines / theorems:** of the folder, in the final project.
- **Checks:** the brute-force check made before sending, with its negative control.

The pilot, piece 1, ran on Aristotle from 10:48 to 11:27 on 28 Sep. It was downloaded and compiled together with piece 2.

#	Tarball	md5	Downloaded	Build (s)	Peak RSS (GB)	Files	Lines	Thms	Brute-force checks before sending
1–2	(base)	fd15dc24	28 Sep 12:39	212.4	2.32	5+4	897+569	26+19	— (the $q = 3$ case; see paper §12)
3	(1)	2961992d	28 Sep 13:26	84.2	2.28	3	574	21	—
4	(2)	cfa67c5f	28 Sep 13:55	120.7	2.20	4	668	37	— (examples checked in Lean by <code>decide</code>)
5	(3)	6553319e	28 Sep 14:22	74.8	2.52	4	431	28	11 541 pairs, 0 failures; the parity control fails, as it should
6	(4)	1d5f698b	28 Sep 15:42	106.5	2.15	3	599	32	10 759 tails, 0 failures
7	(5)	19460305	28 Sep 16:39	103.1	2.16	3	862	48	sympy, $q \in \{3,5,7,9\}$, 0 failures
8	(6)	88367171	28 Sep 17:23	270.3	2.00	6	1137	59	106 inclusions, 0 failures; control (one slice lower): 47 failures
9	(7)	a15d2cc0	28 Sep 17:50	90.5	2.17	2	220	7	35 cases, 0 failures (equality in all)
10	(8)	0b22b201	28 Sep 18:28	265.9	1.99	7	1167	63	8 cells: $\dim = Q_k(q) = I' = \#closed$, 0 failures
11	(9)	4851b4c0	28 Sep 19:15	104.8	2.06	3	658	25	135 random cases, 0 failures
12	(10)	02f9c834	28 Sep 19:57	129.2	2.20	4	708	29	evaluation lemma at every pair of T ; support lemma at 1 192 non-zero evaluations; 0 failures (re-run 29 Sep)
13	(11)	dcbc02c3	28 Sep 21:20	111.0	2.23	4	713	28	160 cases over F_2, F_3, F_5, F_7 , 0 failures
14	(12)	041b7302	28 Sep 21:53	73.3	2.40	3	406	12	integer Smith forms, 19 cases, 0 failures
15	(13)	fd52e181	28 Sep 22:26	97.8	2.58	5	973	35	25 936 points, 0 failures
16	(14)	c9cb3502	28 Sep 23:17	87.3	2.40	4	960	39	5 356 tails + 532 shapes + 3 793 pairs, 0 failures
17	(19)	c6a643fb	29 Sep 06:58	151.4	1.71	2	317	22	6 491 pairs, 0 failures; control (no transfer of sizes): 183/770 fail
18	(20)	7918d41a	29 Sep 07:39	233.7	1.69	3	898	43	244 inclusions, 0 failures; control (one slice lower): 26/48 fail; control ($q = 5$ over F_3): 2/26 fail
19	(21)	5bfc98b6	29 Sep 08:26	116.5	1.67	2	444	23	62 down-sets, 0 failures
20	(22)	a87f28b2	29 Sep 09:29	109.9	1.76	2	419	17	12 cells, 0 failures; control (q not a power of p): the projection is undefined
21	(23)	9aa13322	29 Sep 10:02	152.1	1.74	2	467	30	1 995 checks, 0 failures
22	(24)	890fb2a5	29 Sep 11:13	139.7	1.99	4	653	29	4 643 740 checks, 0 failures; control (r even) fails

#	Tarball	md5	Downloaded	Build (s)	Peak RSS (GB)	Files	Lines	Thms	Brute-force checks before sending
23	(25)	8115bb0d	29 Sep 11:57	131.7	1.97	3	548	24	72 trials, 0 failures; control (overlapping blocks): 12/12 fail
24	(26)	ab44d0e5	29 Sep 12:27	197.6	2.07	4	569	17	43 colourings, 0 failures
25	(27)	270db1c2	29 Sep 13:31	131.1	2.37	5	1104	42	61 checks, 0 failures; control: 19/27 fail
26	(28)	5537efe2	29 Sep 15:56	264.8	1.99	6	1478	71	49 checks, 0 failures; control: 5/5 fail
27	(29)	9546dac4	29 Sep 16:52	195.0	2.08	4	865	25	8 cells, 0 mismatches per colouring; two controls: 8/8 and 8/8 fire
28	(30)	6d17cd42	29 Sep 17:39	239.6	2.15	7	1199	55	6 checks, 0 failures; control ($t + 1$ for $t - 1$): 5/5 fire
29	(31)	9a1892dc	29 Sep 18:35	113.9	2.50	5	623	31	ranks = $Q_k(m)$ over $F_2 \dots F_{13}$ and $F_{1000003}$ in 5 cells; $m = 15$ over F_2, F_3, F_5 ; 546; 0 failures

Notes on the table.

- Tarballs (15)–(18) are duplicates of (14) (same md5 c9cb3502...). They are delayed releases of blocked download clicks, kept and not used.
 - The files and lines of piece 1 include RequestProject/Main.lean (120 lines, 2 theorems).
 - The brute-force scripts of pieces 7–16 and 18–29 are kept (*checks/*); those of pieces 5, 6 and 17 were not kept, and their numbers are the ones recorded in the state file when the pieces were sent. Pieces 1–4 had no separate script.
 - The builds are incremental: each builds only the new folder on top of the compiled project. The clean rebuild of everything is in §8.
-

7. Totals

Quantity	Value
Pieces sent to Aristotle	29 (one task each)
Properties registered by Aristotle	137 PROVED, 0 NEGATED, 0 in progress
Lean files	113
Lines of Lean	21 126
Theorems and lemmas (<i>theorem/lemma</i> at the start of a line)	937
Definitions (<i>def, abbrev, structure, inductive, instance, class</i>)	351
sorry / admit / added axioms / native_decide	0 / 0 / 0 / 0
Axioms of <code>mainTheorem'</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
Calendar time	28 Sep 10:48 (pilot sent) → 29 Sep 18:35 (last piece downloaded): 31 h 47 min, including the night of 28–29 Sep. Work paused between 23:27 and 06:58.
Sum of the 28 incremental local builds	4 108.7 s = 68.5 min
Largest local build	270.3 s (piece 8); peak resident memory 2.58 GB (piece 15)
Clean rebuild of the whole project, one module at a time	113 of 113 modules, all exit 0; 18:41:27 → 20:08:22, i.e. 1 h 26 min 55 s wall-clock (5 214 s inside the 113 <i>lake build</i> calls); slowest module <code>TheoremB.Root</code> , 117 s
Declarations of the project in the dependency cone of <code>mainTheorem'</code>	2 367, in 90 files and 26 folders (§3.2)

8. Reproducibility

Machine: Apple Mac (arm64), 8 CPU cores, 8 GB of memory, macOS (Darwin 25.5.0).

Toolchain: `leanprover/lean4:v4.28.0`; `Mathlib v4.28.0` = `commit 8f9d9cff6b-d728b17a24e163c9402775d9e6a365` (from `lake-manifest.json`).

Commands, from the project root (`output-final_aristotle/`):

```
lake exe cache get      # Mathlib's compiled files
lake build              # all 113 files; exit 0
lake env lean Check.lean # a file with: import RequestProject.ColAssembly.Main
                        #                #print axioms ColAssembly.mainTheorem'
```

Clean rebuild. On 29 Sep 2026 the project's own build directory was moved aside (Mathlib untouched) and all 113 modules were rebuilt one at a time in dependency order, each with its own *lake build*, so that only one Lean process ran at once. The result: all 113 modules built with exit code 0, from 18:41:27 to 20:08:22 (1 h 26 min 55 s), with 5 214 s spent inside the 113 *lake build* calls. The slowest module was `RequestProject.TheoremB.Root`, at 117 s. Afterwards a plain *lake build* reported «Build completed successfully (8139 jobs)», and *#print axioms* on the fresh build gave the same three axioms for all six main theorems. Log: `projecto_lean/clean_rebuild_2026-09-29.log`.

Fingerprints of the source:

- SHA-256 of the concatenation of the 113 .lean files, in sorted path order: 7d26fa1cf203aaa900c0feb-b1fc8a24308b722460b9dbe1d169a32953f357498.
- SHA-256 of the list of their individual SHA-256 lines (shasum -a 256 \$(find RequestProject -name '*.lean' | sort) | shasum -a 256): 81fdc844d7d57a37b9ebb832d7be4c202fc18e8bb-d145615c584325b49445f94.

In the public repository (github.com/tretoef-estrella/chaise-longue-theorem, folder *lean*): *project/* (the Lean project), *pieces/* (the 29 pieces), *checks/* (the brute-force scripts and the Lean check files), *logs/* (all builds, checks, the clean rebuild and the dependency cone), *AUDIT_LOG.md*.

On the author's machine (in the folder *ARISTOTLE_LEAN* of the author's archive):

- proyecto_lean/output-final_aristotle/: the Lean project;
- run3/ ... run29/: every returned project, extracted;
- q*.md: the 29 pieces sent;
- proyecto_lean/build_run*.log, check_run*.log: every build and every check;
- INFORME_ARISTOTLE_v1.md: the audit log, one section per run;
- ESTADO_ARISTOTLE.md: the queue with the brute-force numbers.

9. What this certificate claims, in one paragraph

A computer proof assistant, Lean 4 with the Mathlib library, has checked a complete proof of the following statement, with no gap and no axiom beyond the three standard ones:

For every odd m and every $k \geq 0$, the quotient of the integral group ring $Z[(Z/m)^{2k+1}]$ by the ideal generated by the Degtyarev–Shimada elements ψ_j is a free abelian group of rank $m^{2k+1} - Q_k(m)$, and over every field the corresponding quotient has that dimension.

By [DS, Theorem 1.1(a)] (Pham's theorem and the intersection numbers of [DS, Theorem 2.2], cited and not formalized), this is equivalent to Conjecture 1.2 of Degtyarev and Shimada for the Fermat varieties of odd degree m in even dimension $2k$.

The proofs in Lean were written by the AI system Aristotle (Harmonic) from pieces written by the author. The author's audit, assisted by Claude (Anthropic), checked that the statements match the paper, and compiled and re-checked everything on his own machine.

References

- [DS] A. Degtyarev, I. Shimada, *On the topology of projective subspaces in complex Fermat varieties*, J. Math. Soc. Japan **68** (2016), no. 3, 975–996; doi:10.2969/jmsj/06830975; arXiv:1405.4683v3.
- [Ph] F. Pham, *Formules de Picard–Lefschetz généralisées et ramification des intégrales*, Bull. Soc. Math. France **93** (1965), 333–367.
- R. Amichis Luengo, *The Chaise Longue Theorem*, v7, Zenodo 2026, DOI 10.5281/zenodo.22961151.
- The Lean 4 theorem prover, <https://lean-lang.org>; the Mathlib library, <https://github.com/leanprover-community/mathlib4>.
- Aristotle, Harmonic, <https://aristotle.harmonic.fun>.